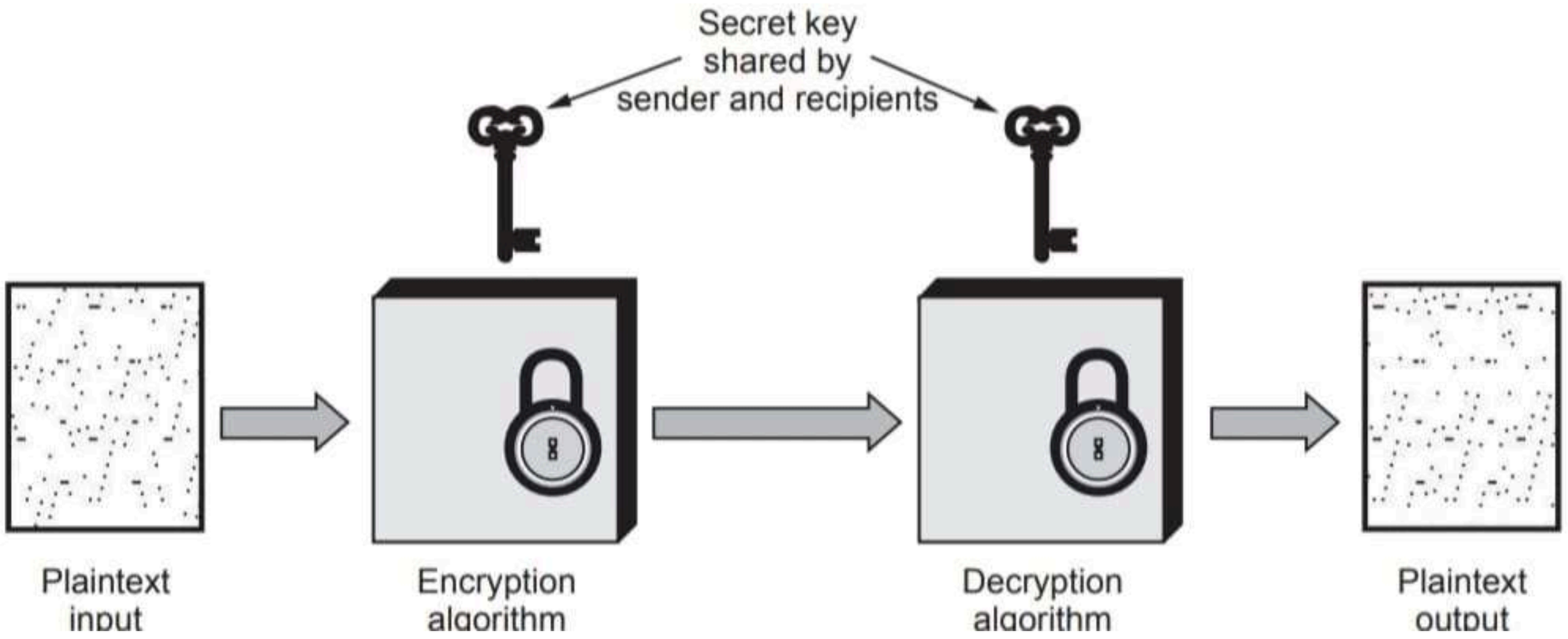
Information Security

Unit 2

Symmetric Key Cryptography

Introduction to symmetric key cryptography



Introduction to symmetric key cryptography

1.Plaintext

- Original readable message or data.
- Input given to encryption process.

2.Encryption Algorithm

- Converts plaintext into coded form.
- Uses mathematical rules and transformations.

3.Secret Key

- Special value used by algorithm.
- Controls how encryption happens.
- Must be kept private and secure.

4.Ciphertext

- Encrypted message.
- Looks like random characters.
- Depends on plaintext + key.

5.Decryption Algorithm

- Converts ciphertext back to plaintext.
- Uses same secret key.

6.Encryption & Decryption

- Encryption = Plaintext → Ciphertext
- Decryption = Ciphertext → Plaintext

Introduction to symmetric key cryptography

Example (Step-by-Step)

1. Message: HELLO
2. Secret Key: 3
3. Method: Shift each letter 3 steps forward

Encryption:

- H → K
- E → H
- L → O
- L → O
- O → R

Ciphertext: KHOOR

Decryption: Shift back 3 → HELLO

Cryptanalysis

- Cryptanalysis is a technique used to decode encrypted messages.
- It converts unreadable ciphertext into readable plaintext.
- Done without knowing the secret key.

Goal of Cryptanalysis

- To break cipher text.
- To obtain original message.
- To test strength of encryption system.

Cryptanalyst

- A person who performs cryptanalysis.
- Studies encryption system.
- Tries to find hidden patterns or weaknesses.

Cryptanalysis

Types of Cryptanalytic Attacks

- Ciphertext-Only Attack
- Known-Plaintext Attack
- Chosen-Plaintext Attack
- Chosen-Ciphertext Attack

Ciphertext-Only Attack

- Attacker has only ciphertext.
- No key and no plaintext.
- Tries to guess original message.

Known-Plaintext Attack

- Attacker knows:
  - Some plaintext
  - Corresponding ciphertex
- Uses this info to find key.

Cryptanalysis

Chosen-Plaintext Attack

- Attacker chooses plaintext.
- Gets its ciphertext.
- Uses patterns to crack system.

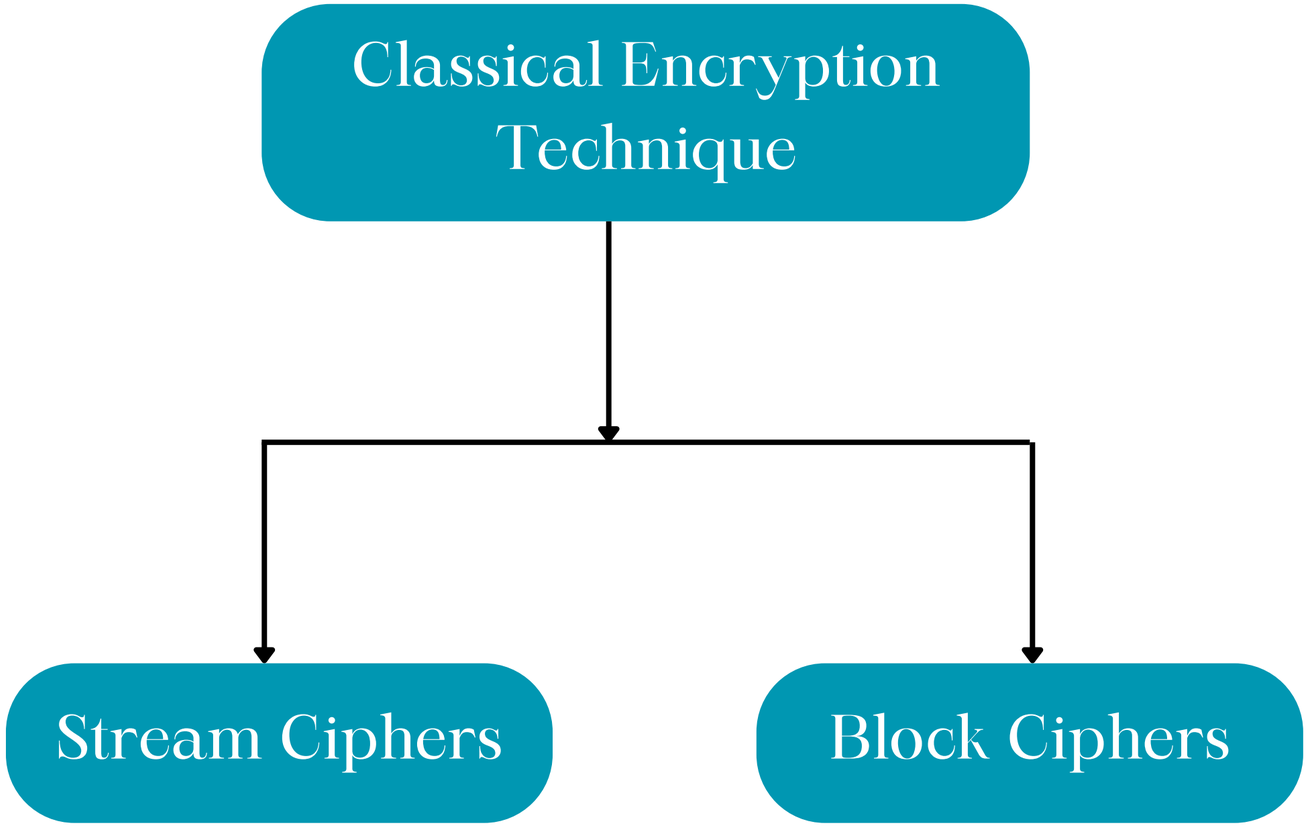
Chosen-Ciphertext Attack

- Attacker chooses ciphertext.
- Gets decrypted plaintext.
- Uses results to discover key.

Cryptanalysis Forms

Linear Cryptanalysis	Differential Cryptanalysis
Uses **linear equations** to describe relationship between plaintext, ciphertext, and key bits.	Uses **differences between plaintext pairs** to analyze how they affect ciphertext differences.
Based on finding **linear approximations** of cipher operations.	Based on studying **input difference → output difference patterns**.
Works by analyzing many plaintext–ciphertext pairs.	Works by analyzing pairs of plaintext with specific differences.
Goal: Find key bits using probability bias in linear relations.	Goal: Find key by observing which input differences produce predictable outputs.
Attack type: **Statistical attack**.	Attack type: **Chosen-plaintext attack**.
Requires large amount of known plaintext data.	Requires chosen plaintext pairs.
Easier against ciphers with weak substitution boxes (S-boxes).	Effective if cipher shows predictable difference propagation.
Example: Attacking a block cipher where equation like $P_1 \oplus C_2 = K_3$ holds with high probability.	**Example:** If two plaintexts differ by 0001 and ciphertext difference often becomes 0100, attacker uses that pattern to guess key.

Classical Encryption Technique



Stream Ciphers

- A stream cipher encrypts data one byte or one bit at a time.
- Decryption also happens one byte at a time.
- Works continuously like a stream.

How It Works

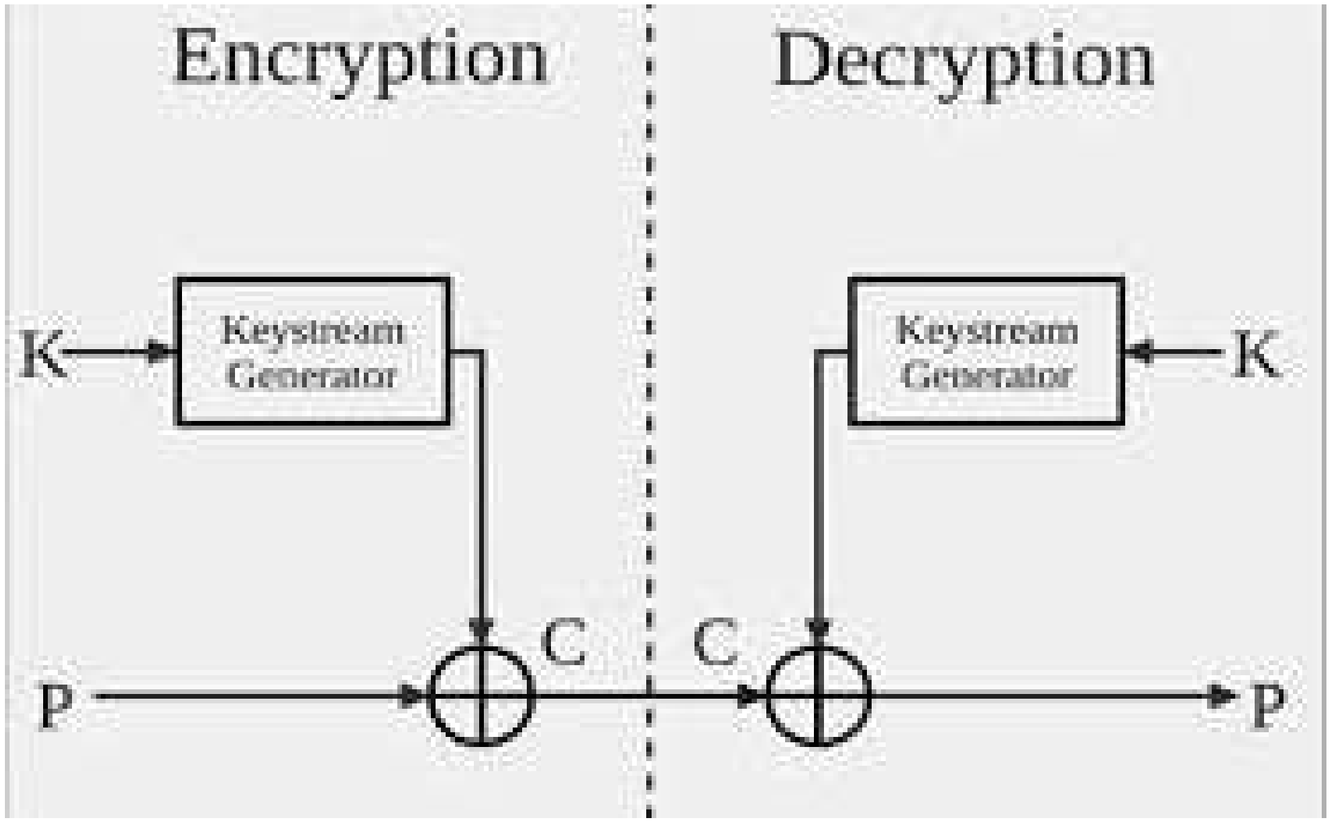
- Uses a Key Stream Generator.
- Generator produces a random-like key stream.
- Plaintext is combined with keystream using XOR operation.

Formula:

Ciphertext = Plaintext ⊕ Keystream

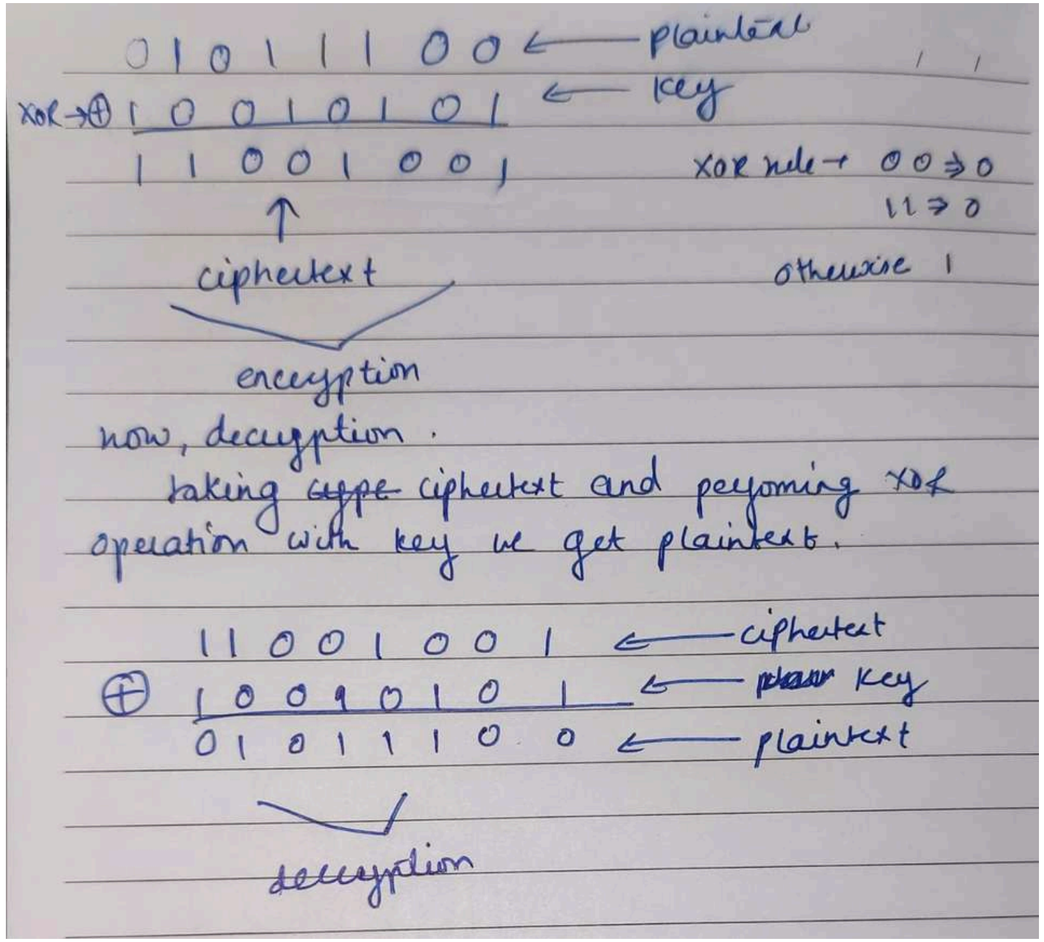
Stream Ciphers

Diagram



Stream Ciphers

- Plaintext (binary): 01011100
- Keystream: 10101001
- XOR Result: 11110101 → Ciphertext



Stream Ciphers

Advantages

- Very fast encryption.
- Low hardware complexity.
- Suitable for real-time communication (audio/video).

Disadvantages

- Low diffusion (small change affects small part).
- If key is reused → security breaks.
- Vulnerable to modification attacks.

Block Ciphers

- Block cipher encrypts data one block of text at a time.
- Each block is processed as a whole unit.
- Decryption also works block by block.

How It Works

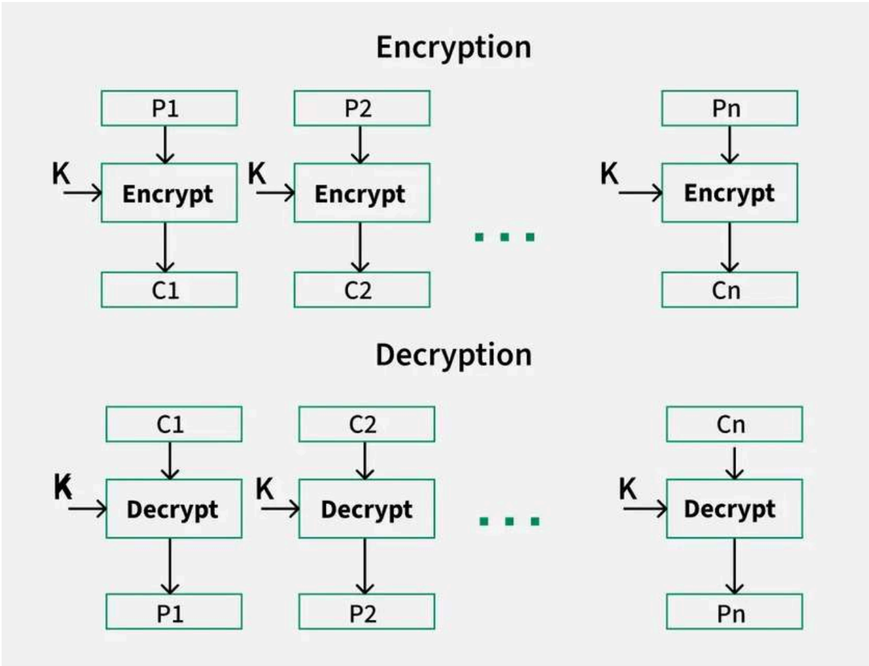
- Plaintext is divided into fixed-size blocks (e.g., 64 bits or 128 bits).
- Each block is encrypted using:
  - Algorithm
  - Secret key
- Output is ciphertext block.

Block Ciphers

Example Flow

- Plaintext Block → Encryption Algorithm + Key → Ciphertext Block
- Ciphertext Block → Decryption + Same Key → Original Plaintext

Diagram



Block Ciphers

Advantages

- Strong encryption.
- Less pattern visibility.
- Suitable for secure file storage and databases.

Disadvantages

- Slower than stream cipher.
- Error propagation:
  - If one bit error occurs,
  - Whole block may be affected.

Difference between stream cipher and block cipher

Feature	Stream Cipher	Block Cipher
Data Processing	Encrypts 1 bit/byte at a time	Encrypts fixed-size block (e.g., 64/128 bits)
Speed	Faster	Slower
Operation	Continuous stream	Block-by-block
Diffusion	Low	High
Error Effect	Error affects only that bit/byte	Error affects whole block
Security	Less secure if key reused	More secure overall
Complexity	Simple algorithm	More complex algorithm
Use Cases	Real-time data (audio, video)	File storage, databases
Example Method	XOR with keystream	Substitution + permutation rounds

Substitution Technique

Substitution Technique

- A substitution cipher replaces each character of plaintext with another character.
- Replacement is done using a rule based on a key.
- Letters are taken from the same alphabet set.

How It Works

- Plaintext letters → replaced → Ciphertext letters
- Each letter maps to another letter.
- Mapping depends on the chosen key.

Caesar Cipher

- Caesar cipher is a special type of substitution cipher.
- Each letter is shifted by a fixed number of positions.
- Shift value acts as the key.

Substitution Technique

Caser Cipher Steps :-

1) First write numerical equivalent to each letter as shown below

- Numerical equivalent to each letter is given below.

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Example :- HELLO

2) Now , for encryption use formula :-

CT = (plain text + key) mod(no.of letters)

and for decryption use formula:-

PT = (CT - Key) mod (no.of letters)

Substitution Technique

- Numerical equivalent to each letter is given below.

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

HELLO → KHOOR

Example :- HELLO

3) Solve by observing the table :-

for encryption use formula :-

CT = (plain text + key) mod(no.of letters)

first letter → H → CT= (7+ 3) mod 26

now, 10 mod 26 =10

by observing the table 10 is represented by K

so, H → K

do , same for other letters and for decryption as well.

Substitution Technique

Advantages

- Very simple to understand.
- Easy to implement.

Disadvantages

- Very weak security.
- Can be easily broken using frequency analysis.

Substitution Technique

Monoalphabetic Cipher

- A substitution cipher where each letter of plaintext maps to a fixed letter.
- Same substitution rule is used throughout message.
- One letter → always replaced by same letter.

Key Idea

- Uses a single substitution alphabet.
- Key is the arrangement of letters used for substitution.
- Total possible keys = 26! (very large).

EXAMPLE :-

Substitution Technique

Monoalphabetic Cipher

Plain	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Cipher	J	I	B	R	K	T	C	N	O	F	Q	Y	G	A	U	Z	H	S	V	W	M	X	L	D	E	P

one permutation of the possible 26!

plain text : c r y p t o g r a p h y

cipher text : B S E Z W U C S J Z N E

Hence we obtain the cipher text as "BSEZWUCSJZNE"

Substitution Technique

Playfair Cipher

- Playfair cipher is a digraph substitution cipher.
- Encrypts letters in pairs (two at a time) instead of single letters.
- Uses a 5×5 matrix of letters based on a keyword.

Key Matrix

- Keyword is written first (without repeating letters).
- Remaining alphabet letters are filled in order.
- Usually I and J are treated as same letter.

Example key: MONARCHY

Substitution Technique

Playfair Cipher

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	O	S	T
U	V	W	X	Z

Encryption Rules

- Split plaintext into letter pairs.
- If both letters same → insert filler letter (usually X).
- If letters in same row → replace with letters to right.
- If same column → replace with letters below.
- If rectangle → swap column positions.

Substitution Technique

Playfair Cipher

H:- (2,2) I (3,4) & D(2,5) E(3,1)

Example

Plaintext: HI DE

Pairs: HI | DE

Using matrix rules →

HI → BF

DE → CK

Ciphertext: BFCK

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	O	S	T
U	V	W	X	Z

Advantages

- More secure than simple substitution cipher.
- Hides single-letter frequency.

Disadvantage

- Still breakable with analysis.
- More complex than Caesar cipher.
- Needs shared keyword.

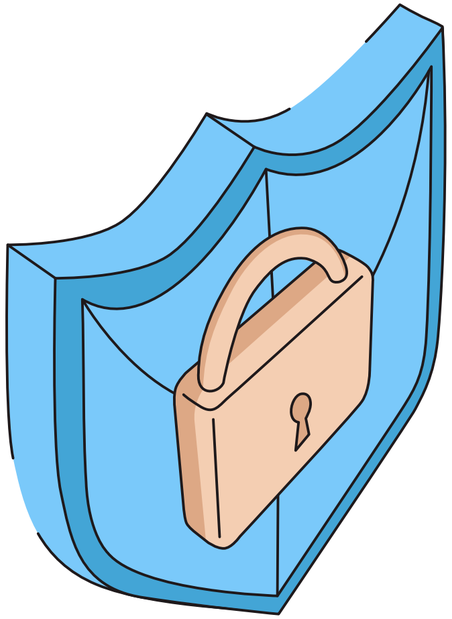
Substitution Technique

Hill Cipher

- Hill cipher is a polyalphabetic substitution cipher.
- Uses matrix multiplication to encrypt text.
- Works on blocks of letters instead of single letters.

Key Concept

- Uses a key matrix (usually 2×2 or 3×3).
- Plaintext letters → converted into numbers:
  - A = 0, B = 1, C = 2 … Z = 25
- Encryption is done using matrix math.



Substitution Technique

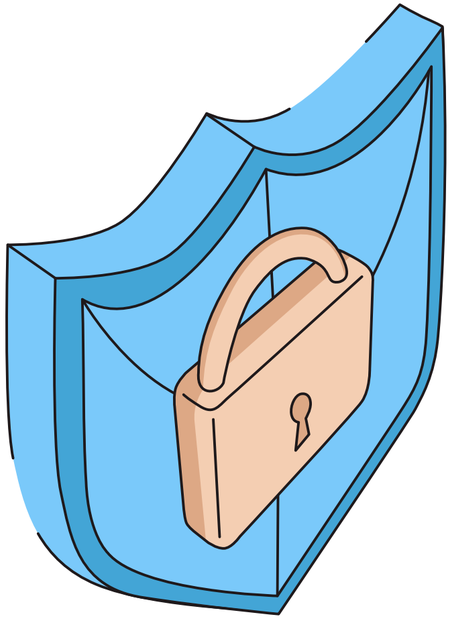
Hill Cipher

Encryption Formula

- $C = K \times P \bmod 26$
- Where:
- C = Ciphertext vector
- K = Key matrix
- P = Plaintext vector

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25



Example :-

Substitution Technique

Example 2.4.3 *Using hill cipher encrypt plain text "COE" use key "ANOTHERBZ".*

SPPU : May-16, (End Sem), Marks 5

Solution : Plain text = COE

Key = ANOTHERBZ

For plaintext COE, here C = 2 O = 14 E = 4

Therefore $P = \begin{pmatrix} 2 \\ 14 \\ 4 \end{pmatrix}$

For key ANOTHERBZ the numbers are 0, 13, 14, 19, 6, 4, 17, 1, 25

The numbers in the matrix form :

$$K = \begin{pmatrix} 0 & 13 & 14 \\ 19 & 6 & 4 \\ 17 & 1 & 25 \end{pmatrix}$$

Ciphertext = (Key X Plaintext) Mod 26

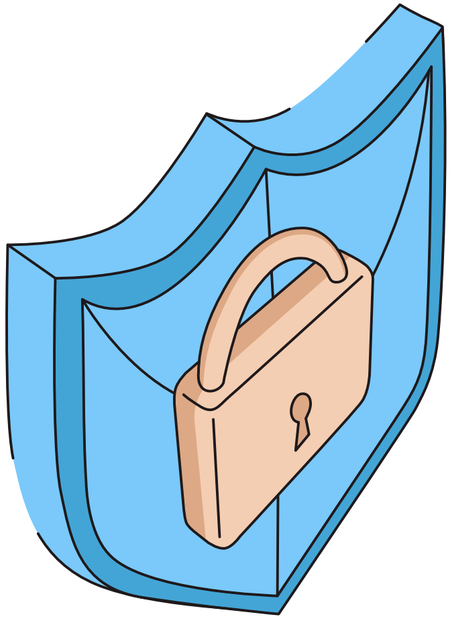
Encryption is as follows :

$$C = \begin{pmatrix} 0 & 13 & 14 \\ 19 & 6 & 4 \\ 17 & 1 & 25 \end{pmatrix} \begin{pmatrix} 2 \\ 14 \\ 4 \end{pmatrix} \bmod 26$$

$$= \begin{pmatrix} 238 \\ 138 \\ 148 \end{pmatrix} \bmod 26 = \begin{pmatrix} 4 \\ 8 \\ 18 \end{pmatrix}$$

Ciphertext := 4 = E , 8 = I and 18 = S

Ciphertext = EIS



Substitution Technique

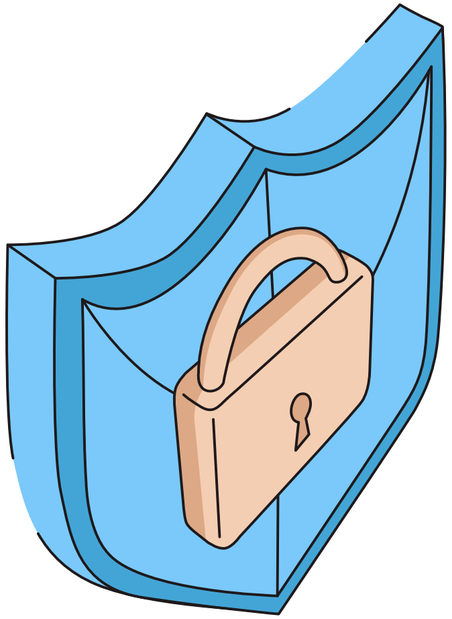
Hill Cipher

Advantages

- Stronger than simple substitution.
- Hides frequency patterns.
- Encrypts multiple letters together.

Disadvantages

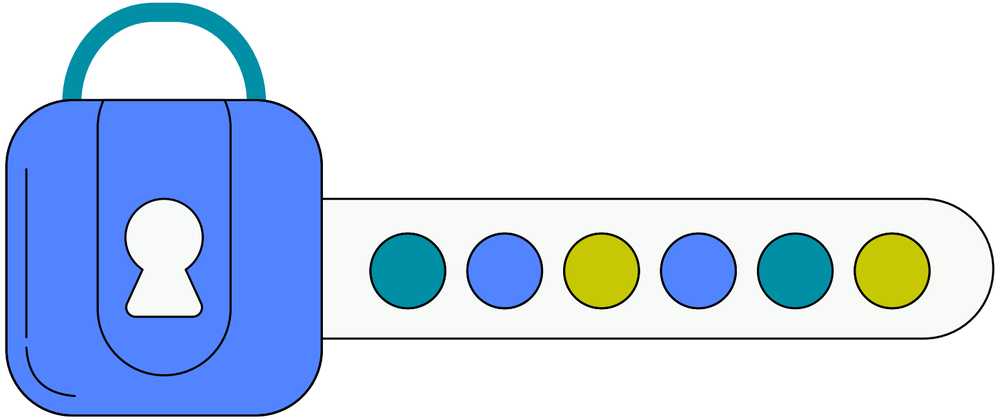
- Requires matrix calculations.
- Key must be invertible.
- If key known → cipher easily broken.



Substitution Technique

One-Time Pad (Vernam Cipher)

- A special type of stream cipher.
- Uses a random key that is same length as plaintext.
- Considered perfectly secure if used correctly.



Basic Idea

- Plaintext is combined with key using XOR operation.
- Each bit/letter encrypted separately.
- Same key is used for decryption.

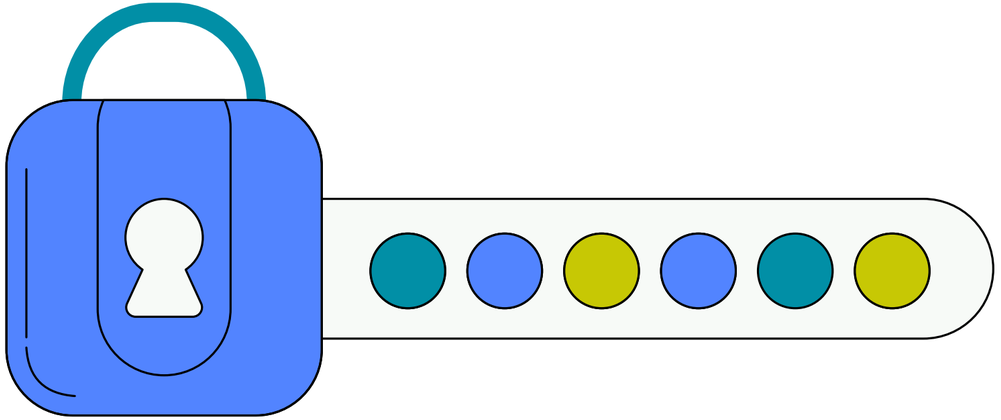
Encryption Formula:-

$$\text{Ciphertext} = \text{Plaintext} \oplus \text{Key}$$

Substitution Technique

One-Time Pad (Vernam Cipher)

Plaintext :	c	o	m	e	t	o	d	a	y
	2	14	12	4	19	14	3	0	24
Key	N	C	B	T	Z	Q	A	R	X
	13	2	1	19	25	16	0	17	23
Total	15	16	13	23	44	30	3	17	47
Subtract 26 if > 25	15	16	13	23	18	04	3	17	21
Ciphertext	P	Q	N	X	S	E	D	R	V



Substitution Technique

Fiestel Cipher

Diagram :-

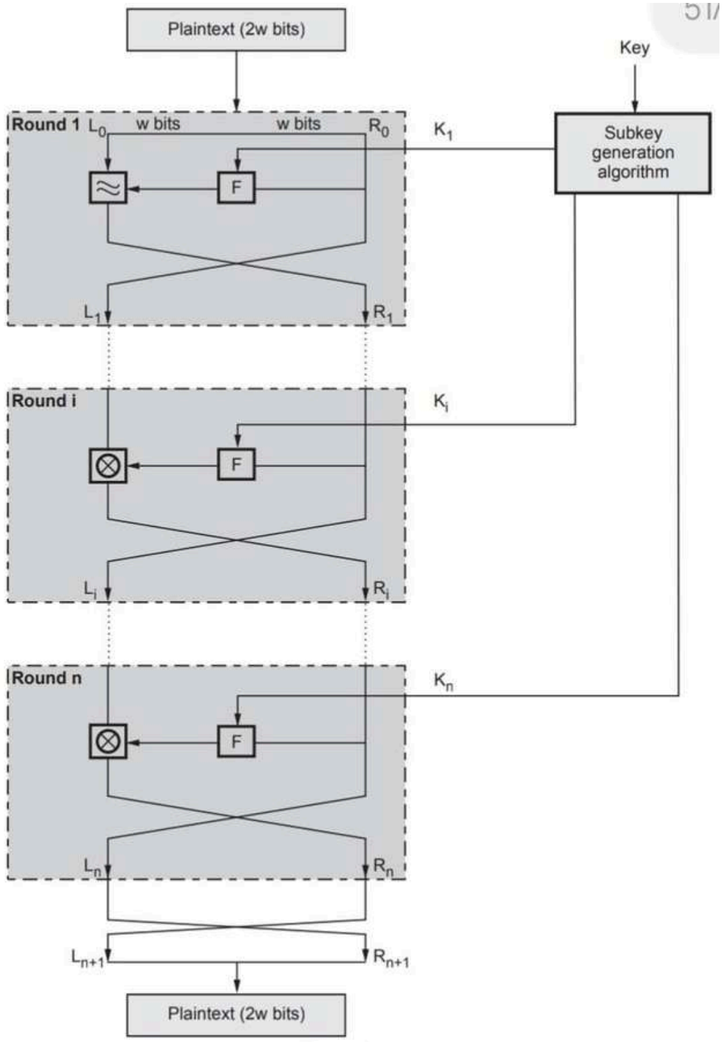
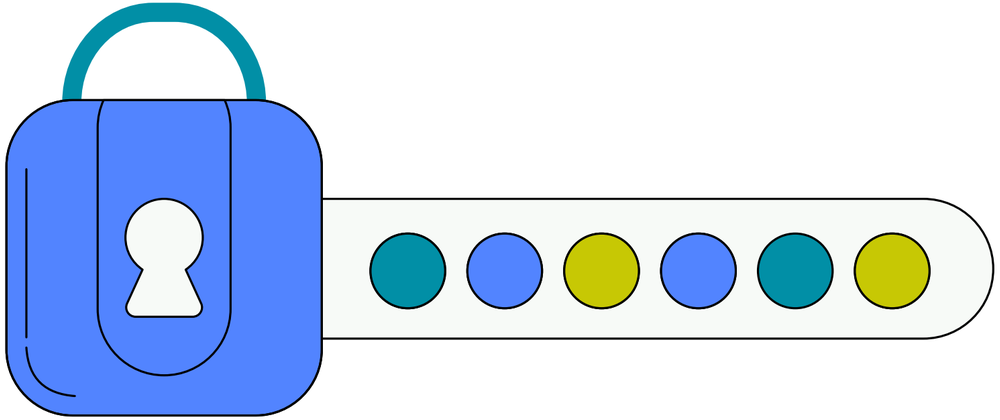


Fig. 2.4.2 Classical feistel network



- $L_i = R_{i-1}$ and
- $R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$

Transposition techniques

- A transposition cipher rearranges the characters in the plaintext to form the ciphertext. The letters are not changed.
- The rail fence cipher is composed by writing the plaintext in two rows, proceeding down, then across and reading the ciphertext across, then down.

Rail Fence Example :-

Plaintext: HELLOWORLD

Write zigzag (2 rails):

H L O O L

E L W R D

Ciphertext: HLOOLELWRD

Transposition techniques

Columnar Transposition Cipher

- A type of transposition cipher.
- Plaintext is written in rows of fixed length.
- Columns are reordered according to a keyword.

Basic Idea

1. Choose a keyword.
2. Write plaintext in rows under keyword letters.
3. Number the keyword letters based on alphabetical order.
4. Read columns in numbered order to get ciphertext.

Transposition techniques

Example

Plaintext: HELLO WORLD

Keyword: ZEBRA

Step 1 — Number the keyword letters

Keyword: Z E B R A

Alphabetical order: A=1, B=2, E=3, R=4, Z=5

Numbered: Z(5) E(3) B(2) R(4) A(1)

Step 2 — Write plaintext in rows under keyword

Z	E	B	R	A
H	E	L	L	O
W	O	R	L	D

Transposition techniques

Step 3 — Read columns in order of numbers 1→5

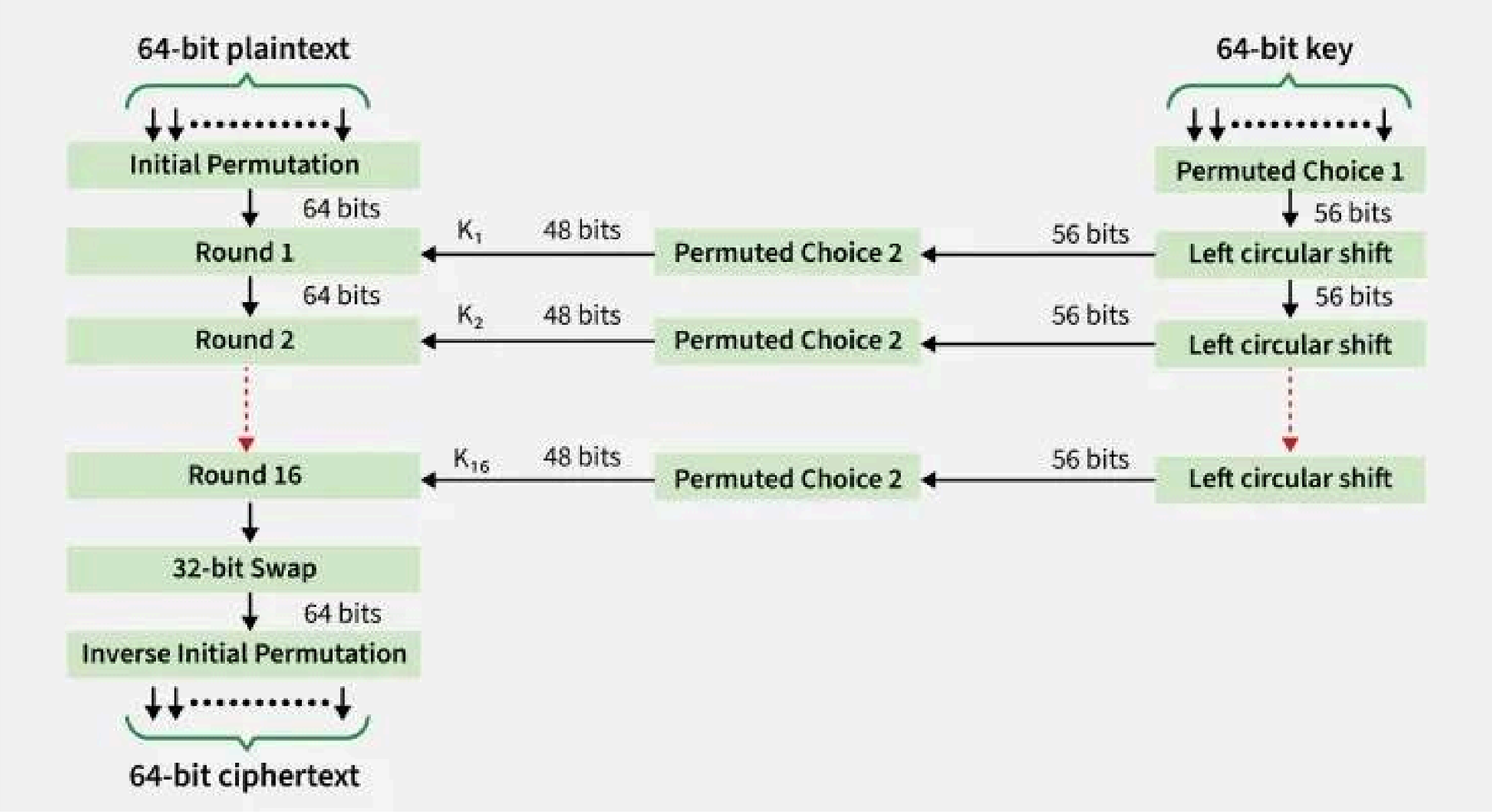
- Column 1 (A) → O D
- Column 2 (B) → L R
- Column 3 (E) → E O
- Column 4 (R) → L L
- Column 5 (Z) → H W

Ciphertext = ODLREOLLHW

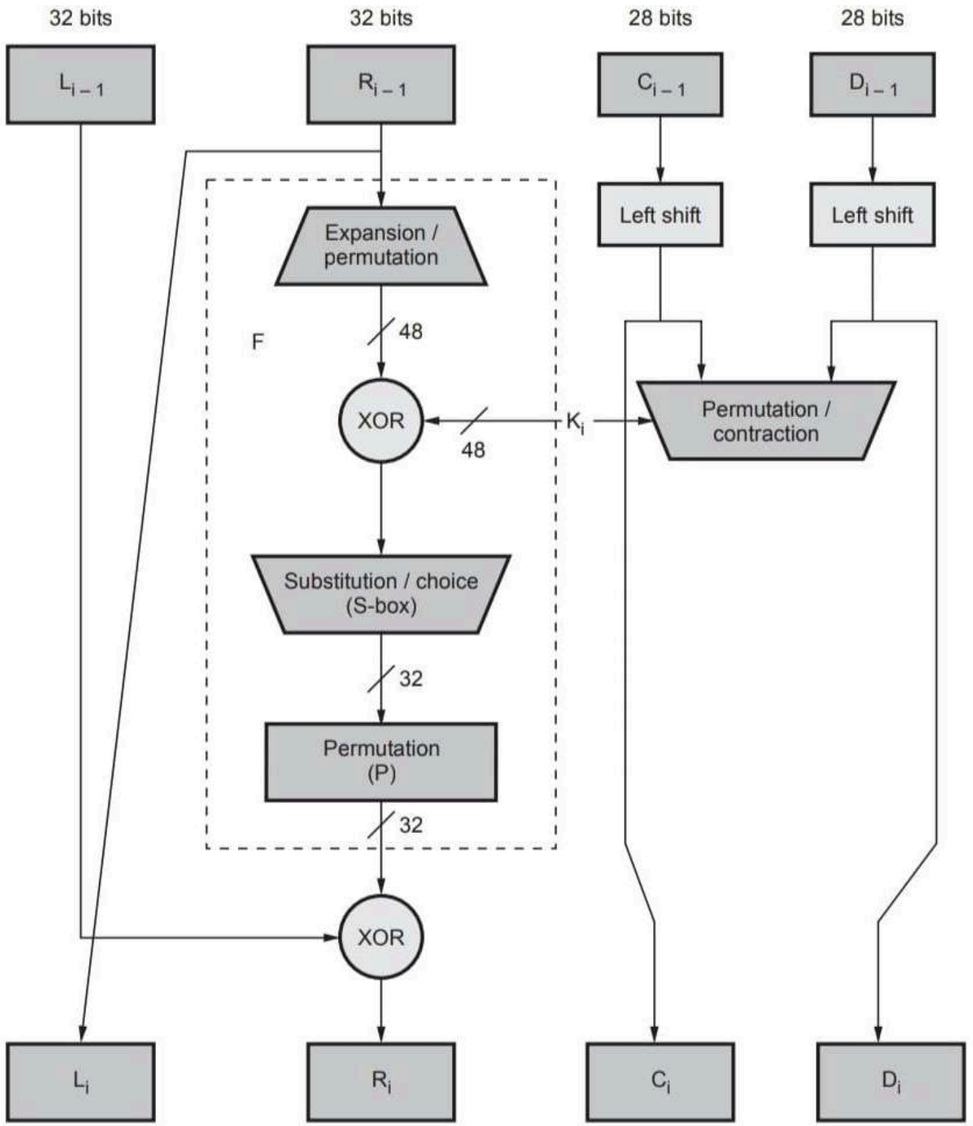
Differences between AES and DES

AES (Advanced Encryption Standard)	DES (Data Encryption Standard)
Block size: 128 bits	Block size: 64 bits
Key size: 128, 192, or 256 bits	Key size: 56 bits
Developed in 2001	Developed in 1977
Based on **Substitution-Permutation** network	Based on **Feistel** structure
Considered highly secure	Vulnerable to brute-force attacks
Faster in software implementations	Slower compared to AES
Used worldwide in modern applications	Mostly obsolete now

Data Encryption Standard



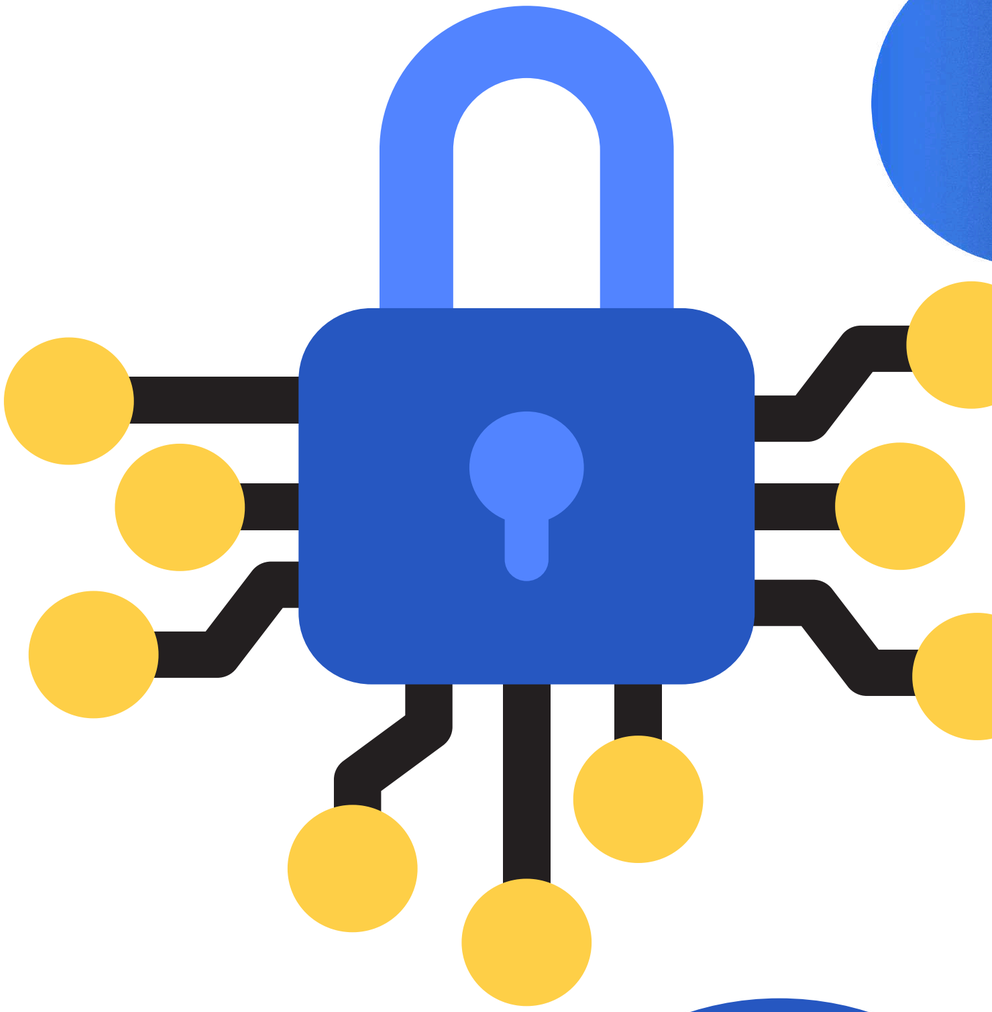
Details of single round



- $L_i = R_{i-1}$ and
- $R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$

Advanced Encryption Standard (AES)

- AES is a symmetric key block cipher.
- Published by NIST in December 2001.
- Used for secure data encryption.
- Evaluation Criteria for AES (NIST)



1. Security

- Measures effort required to cryptanalyse the algorithm.
- Key considerations:
- Actual security compared to other algorithms.
- Randomness – output indistinguishable from random.
- Mathematical soundness of security.
- Other public security concerns during evaluation.

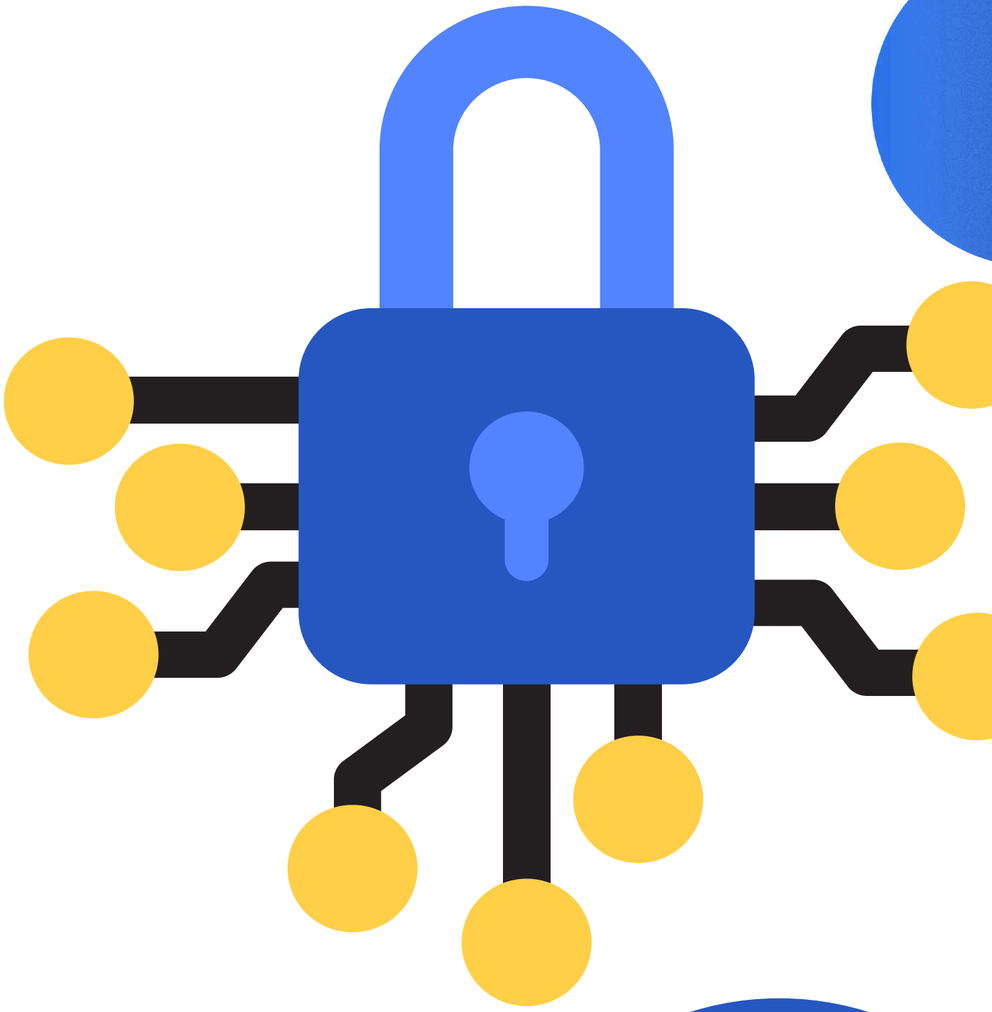
Advanced Encryption Standard (AES)

2. Cost

- Licensing: Worldwide, non-exclusive, royalty-free.
- Computational efficiency: Performance in hardware and software.
- Memory requirements: Considered for both hardware and software implementations.

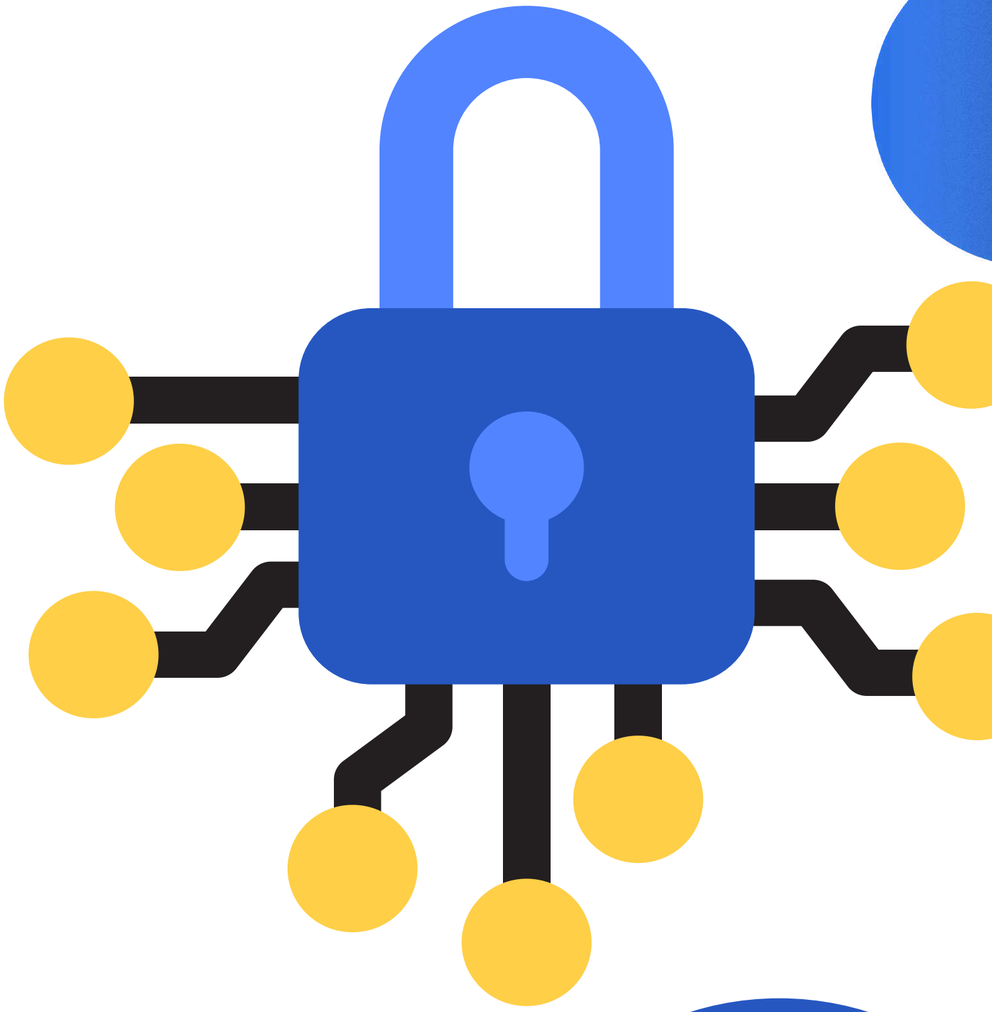
3. Algorithm & Implementation Characteristics

- Flexibility: Suitable for various hardware/software platforms.
- Simplicity: Easier to analyze for security.
- Evaluation considerations:
- General security (public cryptanalysis).
- Software implementations (execution speed, key size variation).



Advanced Encryption Standard (AES)

- Restricted space environments.
- Hardware implementations.
- Resistance to attacks on implementations.
- Balance between encryption & decryption efficiency.
- Key agility (ability to change keys efficiently).
- Versatility & flexibility.
- Potential for instruction-level parallelism.



AES Cipher Overview

- AES is a non-Feistel symmetric block cipher.
- Encrypts/decrypts 128-bit data blocks.
- Key sizes: 128, 192, or 256 bits.

Advanced Encryption Standard (AES)

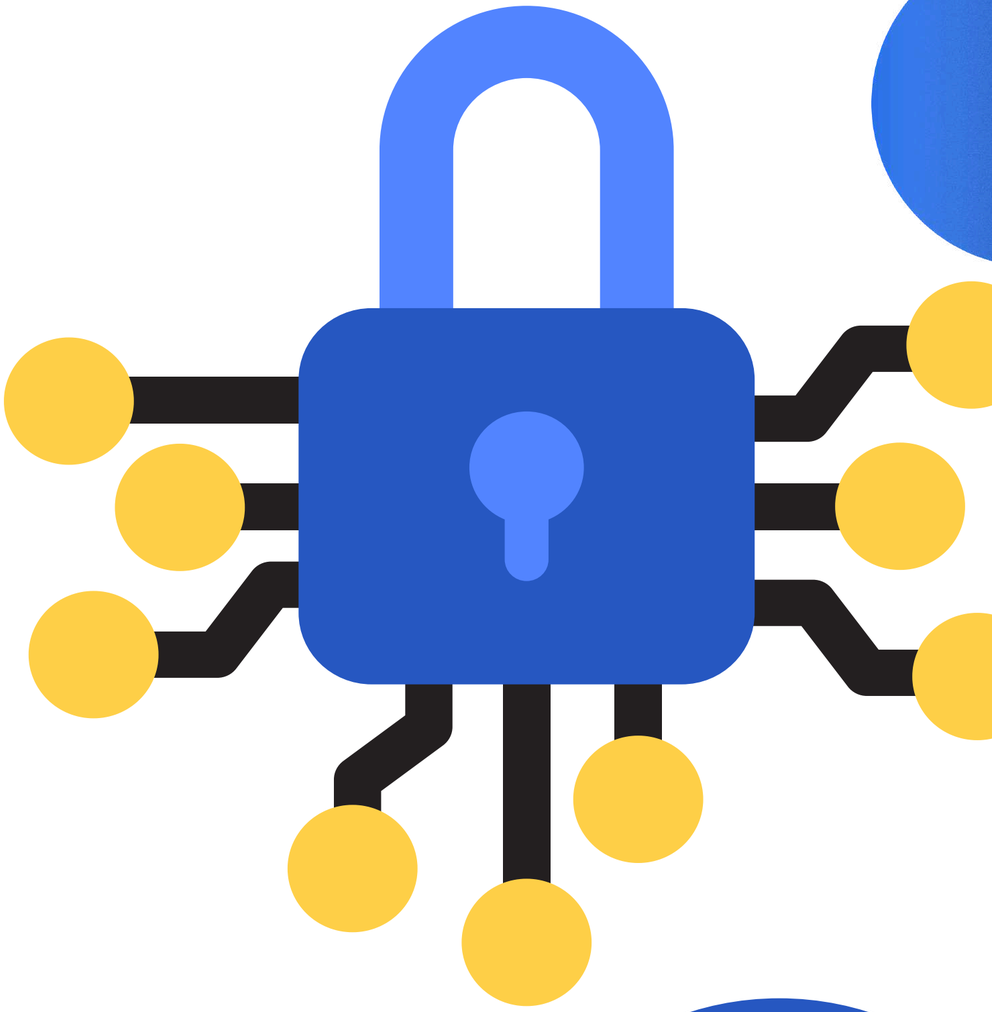
- Number of rounds:
  - 128-bit key → 10 rounds
  - 192-bit key → 12 rounds
  - 256-bit key → 14 rounds

Key Characteristics

- Resistant to all known attacks.
- Fast and compact on multiple platforms.
- Simple design.

AES Round Structure (128-bit key)

Each round has four stages



Advanced Encryption Standard (AES)

- Byte Substitution
- Row Shift
- Column Mixing
- Round Key Addition
- Input block → 128-bit → represented as 16-byte matrix (“state”).
- Data block is transformed through multiple rounds.
- After final stage, state is copied to output.

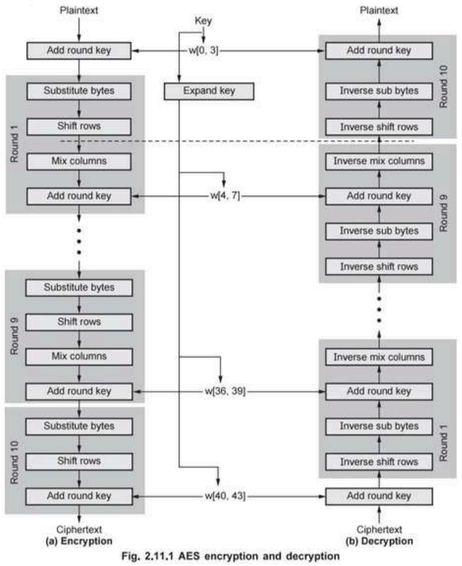
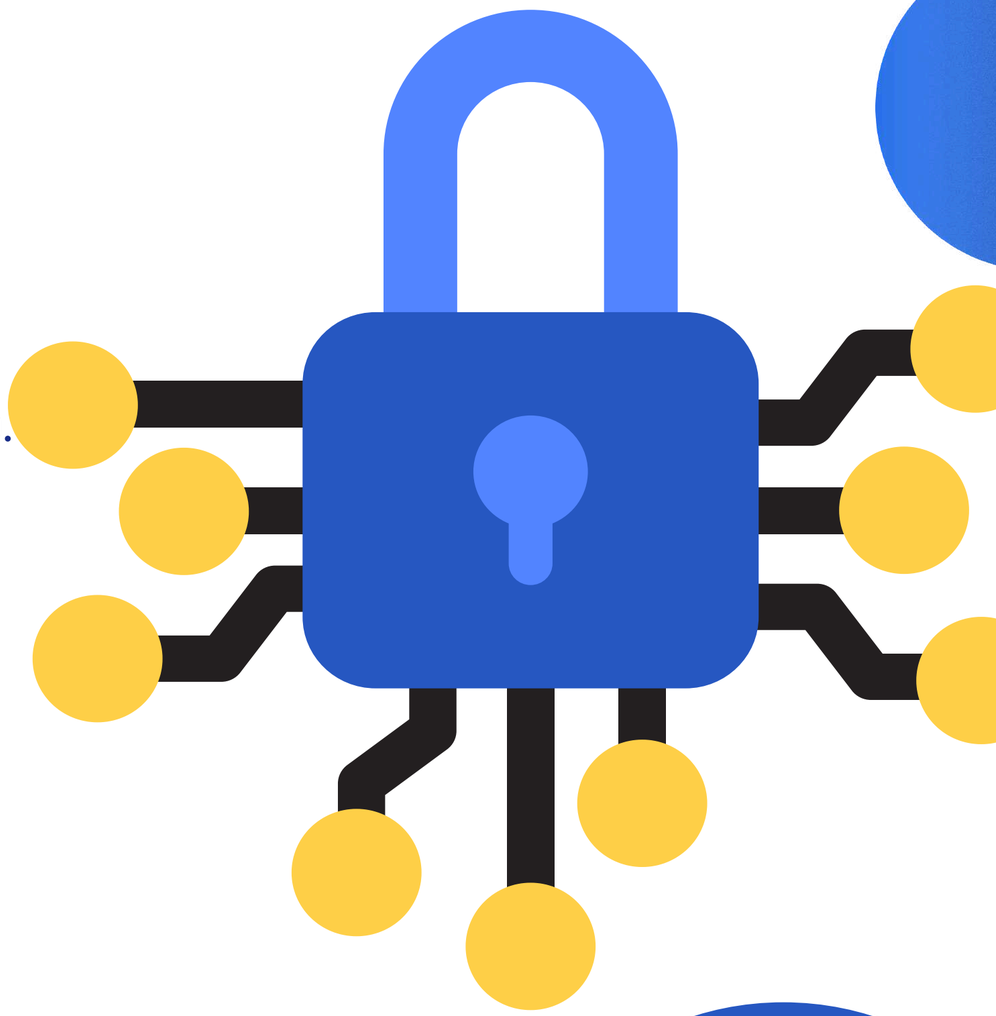


Fig. 2.11.1 AES encryption and decryption



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